

LESSON 3.6

SOLVING TWO-STEP, REAL-WORLD PROBLEMS WITH FRACTIONS OR DECIMALS – PART 1

LESSON INTRODUCTION

MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.

LEARNING TARGETS

- I can solve two-step, real-world problems involving any of the four operations with decimals.
- I can solve two-step, real-world problems involving any of the four operations with fractions.

BENCHMARK COHERENCE

BUILDING ON

MA.5.AR.1 Solve problems involving the four operations with whole numbers and fractions.

WORKING TOWARD

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

ESSENTIAL QUESTIONS

- What are the similarities and differences between solving one-step and two-step real-world problems?
- How can you use visual representations to make sense of decimals and fractions for problem solving?

LESSON NARRATIVE

In this lesson, students apply their knowledge of operations with decimals and fractions to solve various two-step problems. Students use previous understanding from Lessons 1-5 in converting equivalent forms of rational numbers and creating numeric expressions; and solving one-step equations to now solving two-step problems in real-world scenarios. Lessons 7-9 provide additional contexts and practice, so utilize this lesson to center students on the process rather than the product of their work.

COMPONENT SUMMARY

3.6.1 WARM-UP (~5-10 MIN)

Students determine cost of gas per liter

3.6.3 COLLABORATION (20 MIN)

Students collaborate to use visual models and algorithms to solve multi-step problems

3.6.2 GUIDED PRACTICE (20-25 MIN)

Practice solving two-step equations through scaffolded problems

3.6.4 WRAP-UP (~5 MIN)

Confidence scale for solving real-world problems with decimals and fractions

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- N/A

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not round amounts to the nearest cent when solving the Warm-Up problems.
- Students may have difficulty drawing an accurate representation of the ribbon in 3.6.3 Collaboration.

THINGS TO CONSIDER

- Consider reviewing the relationship between money and place value, including how rounding to the nearest cent is the same as rounding to the nearest hundredth.
- Students have additional practice for fluency with decimals and fractions throughout the rest of the unit. Leverage the 3.6.4 Wrap-Up in this lesson as an opportunity to facilitate growth mindset and self-reflection.

LITERACY AND LANGUAGE ROUTINES

Number Talk

3.6.1 Warm-Up orients students with rational numbers with decimals in real-world contexts. When completing and discussing the numbers used, students build computational fluency by reasoning how the numbers are related to the required operations.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

Throughout 3.6.2 Guided Practice and 3.6.3 Collaboration, students select and use the correct operations to solve two-step problems. Consider highlighting when students use flexibility in strategies and various problem-solving approaches, including algorithms connected to visual models.

DIFFERENTIATE INSTRUCTION

Using actual ribbon strands for 3.6.3 Collaboration will activate visual and kinesthetic entry points for representing abstract concepts of fractions and decimals with concrete tools. Students can represent their interpretations of fractional or decimal equivalents using the ribbon.

3.6.1 WARM-UP: GAS GUZZLER

Component Overview: Students use pictures of gas pumps to determine the price of gas per liter and the costs to pump a specific amount of fuel. Consider using Number Talk routines for each question and highlighting student strategies, including estimation or rounding.

TEACHER MOVES

Number Talk

Consider a Number Talk routine to orient students with content involving decimals and the use of unit rate when comparing quantities.

- Are these two different prices for gas? How do you know? (question 1, unit rate)
- Which operation did you use? Why?
- Did you use rounding or estimation? How did you use the decimals for your answer?
- Did you check to see if your answer made sense? How? (feasibility)



3.6.1 Warm-Up: Gas Guzzler

The image below shows the cost of 14.819 and 16.895 liters of gasoline.



1. What is the price per liter of gas?
≈ \$1.37
2. How much will it cost to pump 38 liters of fuel into the tank?
\$52.06



3.6.2 Guided Practice: Two-Step Rational Problems

1. Sophie grew her hair out for a long period of time. Her hair is $2\frac{3}{5}$ feet long. She donated her hair by cutting $\frac{5}{6}$ of a foot off. Sophie's hair then grew $1\frac{2}{3}$ feet.
 - a. What operations are needed to determine the length of Sophie's hair?
Addition and Subtraction
 - b. What values are necessary to determine the length of Sophie's hair?
 $2\frac{3}{5}, \frac{5}{6}, 1\frac{2}{3}$
 - c. Write the numeric expression that can be used to determine the length of Sophie's hair.
 $2\frac{3}{5} - \frac{5}{6} + 1\frac{2}{3}$
 - d. Determine the length of Sophie's hair.
 $3\frac{13}{30}$ ft
 - e. Determine if the length of Sophie's hair makes sense. Support your answer.
Answers will vary. No, the length of Sophie's hair does not make sense. It seems too long.
2. Tiffany has a balance of \$167.32 in her checking account. She deposits \$58.16 in her checking account and then writes a check for \$97.27.
 - a. What operations are needed to determine the amount of money in Tiffany's bank account?
Addition and Subtraction
 - b. What values are necessary to determine the amount of money in Tiffany's bank account?
\$167.32, \$58.16, \$97.27
 - c. Write the numeric expression that can be used to determine the amount of money in Tiffany's bank account.
\$167.32 + \$58.16 - \$97.27
 - d. Determine the amount of money Tiffany has left in her checking account.
\$128.21

3.6.2 GUIDED PRACTICE: TWO-STEP RATIONAL PROBLEMS

Component Overview: Students practice solving two-step equations through scaffolded problems. Consider having students work in small groups or partnerships to engage in math discussions and check work for accuracy for question 1, then work individually to complete question 2. After question 1, engage in class discussion on strategies used and compare strategies for effectiveness.

DIFFERENTIATE INSTRUCTION

Notice the inherently scaffolded prompts in this Guided Practice. While monitoring students' practice, consider asking probing questions as informal observational data for Lesson 7 following parts of question 1.

- 1a: How do you know? What will that look like for letter C?
- 1b: What do you notice about the fractions? What does that make you think about when adding and subtracting those values?
- 1d: Does that mean your answer is wrong? Why or why not?

DIFFERENTIATE INSTRUCTION

Consider providing grid or graph paper for students to organize their work when calculating with decimals.

TEACHER MOVES

Though these problems involve fluency with decimal and fraction operations, position students to focus on the lesson's learning targets as they complete these problems individually. To facilitate this focus, ask questions like:

- How do you know what operation is required?
- How do you know how many steps are involved in finding the final answer?
- How can you check your answer for feasibility within the given context?

3.6.3 COLLABORATION: PROBLEM SOLVING

Component Overview: Students collaborate to solve a problem involving measurements of ribbons. Students draw a visual model to represent and solve two-step numeric problems. Consider placing students in small groups of 3 to 4 to provide support and allow for discussion of various problem-solving strategies and visual models.

DIFFERENTIATE INSTRUCTION

Consider providing ribbon as a manipulative for this activity for visual or kinesthetic learners.

ENRICHMENT

Consider asking scaffolding questions to help students plan and draw their representation, including:

- How can you represent the total length cut into 9 pieces?
- Are the pieces of equal length? How can you show that in your representation?
- What value is unknown or being solved for? How is that visible on your representation?

QUESTIONING

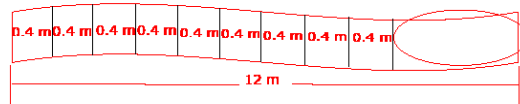
Challenge students to depict the fractional equivalent to their findings. Prompt them with questions like, “Will using fractions instead of decimals change the answer? How does it change your drawing of a ribbon? Which makes more sense to use in this case? Why?” These questions will position students to continually make decisions in Lessons 7-9 when dealing with decimals and fractions in real-world situations.



3.6.3 Collaboration: Problem Solving

1. A roll of ribbon was 12 meters long. Diego cut 9 pieces of ribbon that were 0.4 meters each to tie some presents. He then used the remaining ribbon to make some wreaths. Each wreath requires 0.6 meters.

a. Draw a representation of the information.



b. How does your drawing compare to your partner's drawing?

Answers will vary. The partner's ribbon drawing could be longer. The partner could have an extra section or be missing a section.

c. Draw a circle around the piece of ribbon in your drawing that is available to make wreaths.

d. Write the numeric expression that can be used to determine the amount of ribbon, in meters, available for making wreaths.

$12 - (9 \cdot 0.4)$

e. How many meters of ribbon are available for making wreaths?

8.4

f. What piece of information is not needed to determine the amount of ribbon available for making wreaths?

Each wreath requires 0.6 meters of ribbon.

g. Write a question that uses this information.

How many wreaths can be made from the remaining ribbon?



3.6.4 Wrap-Up: Reflection

Place an **X** on the line to indicate your understanding of each of today's learning targets. Then, provide a brief explanation or evidence to support your self-assessment.

1. I can solve real-world problems with two steps using decimals.

Answers will vary.

I need help. I can't get started.	I'm getting there.	I understand and have evidence.

2. I can solve real-world problems with two steps using fractions.

Answers will vary.

I need help. I can't get started.	I'm getting there.	I understand and have evidence.

3.6.4 WRAP-UP: REFLECTION

Component Overview: Students rate confidence with solving real-world, two-step word problems. Student responses and class observations collected from the lesson combined with this reflection to assess student understanding for small group intervention.

DIFFERENTIATE INSTRUCTION

Consider scaffolded prompts for students to more accurately reflect on their progress.

- I need help: "I don't know where to start" or "Decimals and fractions are hard"
- I'm getting there: "I get some problems right and some wrong" or "I know how to solve but sometimes struggle with decimals"
- I understand and have evidence: "I could explain how to do this to someone else" or "Provide an example to show evidence"